

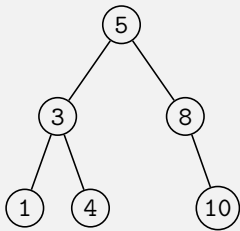
Find Distance in a Binary Search Tree

The distance between two nodes in a tree is defined as the number of edges on the path between those two nodes.

You are given the root of a binary search tree, and two integers p and q . Compute the distance between the nodes with values p and q in the binary search tree.

Example Test Cases:

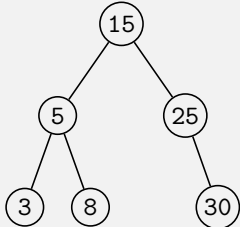
Test Case 1

Input:  , $p = 1, q = 10$

Output: 4

Explanation: There are 4 edges on the path between 1 and 10.

Test Case 2

Input:  , $p = 8, q = 25$

Output: 3

Explanation: There are 3 edges on the path between 8 and 25.

Expected Time Complexity: $O(h)$

(where h is the height of the binary search tree)

Input Format:

The root of a binary search tree, and two integers p and q .

Output Format:

Return a single integer, the distance between the two corresponding nodes.

Constraints:

1. The number of nodes in the binary search tree is in the range $[1, 10^5]$.
2. All node values in the binary search tree are in the range $[1, 10^9]$.

3. All node values in the binary search tree are unique.
4. p and q are present in the binary search tree.

Instructions:

You have been provided with two template files, containing the `main` and `solve` functions respectively. The `main` function, which has been completed for you, parses the input for each test case. For each test case, it calls the `solve` function, which must compute the distance between p and q .

The `solve` function must compute this distance in $O(h)$ time.

(Refer to the `Instructions.pdf` file for detailed guidelines on the input-output format and additional test cases.)

Library Usage

For this lab, you are permitted to use any modules which are part of the standard library of either C++ or python. For C++, do not add any extra `#include` directives; the provided `bits/stdc++.h` covers all allowed modules.